

Extracted Features In Model

No.	Quantity	NPY stem
Molecular structure		
1	Atomic Cartesian positions	pos
2	Covalent bond table	bonds
3	Residue table and sequence links	residues
4	Atom-to-residue index	residue_index
5	Atom chemical-category table	atoms_category_info
6	Enriched atom role	enrichment_role
7	Enriched hybridisation class	enrichment_hybridisation_class
Geometry and secondary structure		
26	N-C-alpha-C angle	tau_N_CA_C
27	N-C-alpha-C validity	tau_N_CA_C_valid
28	N-C-alpha-C-beta angle	angle_N_CA_CB
29	N-C-alpha-C-beta validity	angle_N_CA_CB_valid
30	C-beta-C-alpha-C angle	angle_CB_CA_C
31	C-beta-C-alpha-C validity	angle_CB_CA_C_valid
32	Previous-C-N-C-alpha angle	angle_Cprev_N_CA
33	Previous-C-N-C-alpha validity	angle_Cprev_N_CA_valid
34	C-alpha-C-next-N angle	angle_CA_C_Nnext
35	C-alpha-C-next-N validity	angle_CA_C_Nnext_valid
36	Backbone and side-chain torsion cosines	dssp_torsion_cos
37	Torsion validity mask	dssp_torsion_valid
38	Backbone and side-chain torsion sines	dssp_torsion_sin
39	Peptide omega cosine	omega_cos
40	Peptide omega validity	omega_valid
41	Peptide omega sine	omega_sin
42	X-Pro peptide-bond flag	omega_is_xpro
43	Atomic pyramidalisation	pyramidalization
44	Pyramidalisation validity	pyramidalization_valid
45	DSSP eight-state assignment	dssp_ss8
46	DSSP hydrogen-bond energies	dssp_hbond_energy
47	DSSP observation mask	dssp_observed
48	DSSP polyproline-II quantity	dssp_ppii

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AIMNet2		
8	AIMNet2 per-atom embedding	aimnet2_aim
MOPAC atomic and electrostatic		
9	Coulson atomic charge	mopac_charges_full_precision
10	Atomic bond valency	mopac_bond_valencies_full_precision
11	Atomic s-shell population	mopac_atom_s_population
12	Atomic p-shell population	mopac_atom_p_population
13	Atomic d-shell population	mopac_atom_d_population
14	Lewis-bond partner count	mopac_lewis_bond_count
15	Total Coulson-charge electric field	mopac_coulomb_E
16	Total Coulson-charge field gradient	mopac_coulomb_efg
17	Backbone-source Coulson field	mopac_coulomb_E_backbone
18	Side-chain-source Coulson field	mopac_coulomb_E_sidechain
19	Aromatic-source Coulson field	mopac_coulomb_E_aromatic
20	Backbone-source Coulson field gradient	mopac_coulomb_efg_backbone
21	Side-chain-source Coulson field gradient	mopac_coulomb_efg_sidechain
22	Aromatic-source Coulson field gradient	mopac_coulomb_efg_aromatic
23	Coulson-field scalar summaries	mopac_coulomb_scalars
24	Aromatic field projected along H-parent bond	mopac_coulomb_aromatic_E_proj
25	Aromatic source-atom count	mopac_coulomb_aromatic_n_src
Fixed McConnell responses		
49	Fixed peptide C-N response	mc_peptide_cn_fixed
50	Fixed other-backbone response	mc_backbone_other_fixed
51	Fixed other-side-chain response	mc_sidechain_other_fixed
52	Fixed backbone X-H response	mc_backbone_xh_fixed
53	Fixed side-chain X-H response	mc_sidechain_xh_fixed
Wiberg-weighted responses		
54	Wiberg-weighted peptide C=O response	mc_peptide_co_bo
55	Wiberg-weighted other-backbone response	mc_backbone_other_bo
56	Wiberg-weighted side-chain C=O response	mc_sidechain_co_bo
57	Wiberg-weighted other-side-chain response	mc_sidechain_other_bo
58	Wiberg-weighted disulfide response	mc_disulfide_bo
59	Wiberg-weighted aromatic response	mc_aromatic_bo
60	Wiberg-weighted backbone X-H response	mc_backbone_xh_bo
61	Wiberg-weighted side-chain X-H response	mc_sidechain_xh_bo
62	Wiberg-weighted S-H response	mc_s_h_bo

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Near-field MOPAC responses		
63	Accepted and rejected near-field counts	mc_nearfield_counts
64	Nearest peptide-carbonyl distance	mopac_mc_nearest_co_dist
65	Nearest peptide C-N distance	mopac_mc_nearest_cn_dist
66	Nearest peptide-carbonyl direction	mc_nearest_co_dir
67	Nearest peptide-carbonyl response tensor	mopac_mc_nearest_co_T2
68	Nearest peptide C-N response tensor	mopac_mc_nearest_cn_T2
MOPAC density and orbitals		
69	Atomic-orbital block width per atom	mopac_ao_orbitals_per_atom
70	Atomic AO-density matrix	mopac_atom_ao_density
71	Full-precision Wiberg pair table	mopac_bond_orders_full_precision
72	AO-density pair index table	mopac_bond_density_pairs
73	Pair AO-density blocks	mopac_bond_ao_density
74	Lewis-bond partner indices	mopac_lewis_bond_atoms
75	Localised-orbital energy levels	mopac_lmo_energy_levels
76	Occupied-LMO atom counts	mopac_lmo_occupied_atom_counts
77	Occupied-LMO atom indices	mopac_lmo_occupied_atoms
78	Occupied-LMO coefficients	mopac_lmo_occupied_coefficients
79	Virtual-LMO atom counts	mopac_lmo_virtual_atom_counts
80	Virtual-LMO atom indices	mopac_lmo_virtual_atoms
81	Virtual-LMO coefficients	mopac_lmo_virtual_coefficients
82	Heat of formation	mopac_heat_kcal_mol
83	Point-charge dipole	mopac_dipole_point_charge_debye
84	Hybridisation dipole	mopac_dipole_hybridization_debye
Response derivation support		
85	Topology-aligned MOPAC bond orders	mopac_topology_bond_orders_full
86	Ring inventory	rings
87	Ring atom membership and order	ring_membership
88	Aromatic-ring centres and normals	ring_geometry